Speciation times, refined

Correlation-weighted theory, what "under-confident" means, and removing the classifier error

29 September 2026

This note follows `speciation_explained.pdf`. It answers three questions:

1. Can the profile theory be made more precise by weighting each revealed site's evidence with its correlations to the other revealed sites, using the n-th root of $\prod(1-\rho)$ over its n correlated partners? (§1)
2. What exactly does "the network is under-confident" mean, and how large is the effect? (§2)
3. Can we remove the error of the clade classifier, which is largest at the clade level? (§3)

Where things are:

- code: `scripts/s14_regimes_refined.py` and new functions in `src/psg/regimes.py`;
- tests: `tests/test_18_regimes_refined.py` (8 tests, all passing);
- numbers: `evaluations/md_unconditional/regimes_refined*.json`.

Summary

- **Classifier error was the main reason the clade speciation looked late.** Labelling clones with the exact Bayes posterior of the true mixture moves the network's clade k_S from 15.7 to **10.1** revealed sites. Family and subfamily barely move.
- **The network is under-confident, but only slightly.** Its class distribution behaves like the true one shrunk towards uniform by a factor $\lambda \approx 0.95$ at the clade level (≈ 0.98 subfamily, ≈ 1 family). Correcting for this gives clade $k_S \approx$ **8.2**.
- **Your correlation weighting improves the theory but cannot make it exact.** Within a class almost every pair of sites is correlated, so the geometric mean acts as a roughly uniform discount ($w \approx 0.71$–0.77). At clade level it cuts the calibration error by a third and gives $k_S = 9.7$, close to the network. No single-site weighting calibrates the theory at every k.
- Best estimates of the speciation points: family ≈ 2, subfamily ≈ 5, clade ≈ 8 revealed sites, i.e. $t_S \approx 0.97,\ 0.92,\ 0.86$.

1 Weighting the log-sum by the correlations

1.1 The proposal

The profile log-odds between classes c and c' is a sum of independent-looking terms:

$$\Lambda = \sum_{i \in U} r_i(x^i), \qquad r_i(a) = \log \frac{f_{c,i}(a)}{f_{c',i}(a)}.$$

If sites i and j are correlated within a class, the terms r_i and r_j partly repeat the same evidence, and the sum counts it twice. The proposal is to multiply each term by a weight that shrinks when i has correlated partners among the *revealed* sites:

$$\Lambda_w = \sum_{i \in U} w_i \, r_i(x^i), \qquad w_i = \Big[\prod_{j \in N_i(U)} (1 - \rho_{ij}) \Big]^{1/n_i}, \qquad n_i = |N_i(U)|, \tag{1}$$

where $N_i(U)$ is the set of revealed sites $j \neq i$ that are correlated with i, and $w_i = 1$ if there is none. Equivalently, w_i is the geometric mean of the $(1 - \rho_{ij})$ over those partners. We apply the same weights to the log-likelihood of every class, so the result is still a proper posterior: $\log Q(c \mid x_U) = \log \pi_c + \sum_{i \in U} w_i \log f_{c,i}(x^i) - \text{normaliser}$.

1.2 Which correlation ρ_{ij}?

Residues are categorical, so we use the mutual information and turn it into a number between 0 and 1.

1. **Within-class mutual information.** For each class c at the level considered, with the pair frequencies $f^c_{ij}(a, b)$ of its training sequences:

$$\mathrm{MI}^c_{ij} = \sum_{a,b} f^c_{ij}(a, b) \log \frac{f^c_{ij}(a, b)}{f^c_i(a)\, f^c_j(b)},$$

   averaged over classes with weights equal to their sizes. *Within* the class matters: the class label itself also correlates sites, and that part is the signal we want, not redundancy.
2. **Bias correction.** With a finite sample, independent sites still show MI > 0 (about $(q - 1)^2/2N \approx 0.02$ nats for $N = 8000$ and $q = 20$). We shuffle every column independently inside each class, which destroys all correlations, recompute the MI, and subtract it. A second shuffle gives the noise level σ. A pair counts as correlated (a "partner") if its corrected MI exceeds 5σ.
3. **From MI to a correlation.** Linfoot's informational coefficient of correlation,

$$\rho_{ij} = \sqrt{1 - e^{-2\,\mathrm{MI}_{ij}}} \ \in [0, 1),$$

   is exactly $|r|$ (the Pearson correlation) when the two variables are Gaussian. It is 0 for independent sites and tends to 1 for copies.

(Tested: a copied 4-state column gives MI $= \log 4$ and $\rho = \sqrt{1 - 1/16}$; an independent column gives corrected MI ≈ 0.)

1.3 What the right weight would be

To judge (1), take the simplest case where the answer is known exactly. Suppose the evidence terms of the revealed sites are jointly Gaussian with the same covariance in both classes, mean $\pm \boldsymbol{m}/2$ and correlation matrix R (unit variances for simplicity). The exact log-likelihood ratio of the vector $\boldsymbol{e} = (r_i)_{i \in U}$ is

$$\Lambda_{\text{exact}} = \boldsymbol{m}^\top R^{-1} \boldsymbol{e}.$$

With independent sites, $R = I$ and every term gets weight 1. With correlations, the weights are the entries of $R^{-1}\mathbf{1}$ (for equally informative sites).

- **Two sites with correlation r:** $R^{-1}\mathbf{1} = \frac{1}{1-r^2} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{1+r} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, so each weight is exactly $1/(1 + r)$.
- **$n + 1$ sites, all correlated r with each other:** each weight is $1/(1 + nr)$.
- **General small correlations:** $R^{-1} \approx I - (R - I)$, so $w_i \approx 1 - \sum_j r_{ij}$.

Compare the three simple formulas to first order in the correlations:

weight	definition	first order
geometric mean (your proposal)	$\left[\prod_j (1 - \rho_{ij})\right]^{1/n_i}$	$1 - \frac{1}{n_i} \sum_j \rho_{ij}$ (the *mean* ρ)
product	$\prod_j (1 - \rho_{ij})$	$1 - \sum_j \rho_{ij}$
effective number	$1/\left(1 + \sum_j \rho_{ij}\right)$	$1 - \sum_j \rho_{ij}$
exact (Gaussian)	$(R^{-1}\mathbf{1})_i$	$1 - \sum_j \rho_{ij}$

So for *one* partner your formula is right to first order ($1-\rho \approx 1/(1+\rho)$). With several partners, the n-th root turns the sum into an average, and the discount no longer grows with the number of correlated partners. The product and the effective number have the right first-order behaviour. One more property matters: for two perfect copies, the product and the geometric mean both give weight ≈ 0 to *both*, so the evidence disappears, whereas the effective number gives $1/2$ each, which is right.

A caveat applies to all three: the correlation that matters is that of the evidence terms r_i, not of the residues. r_i is a function of x^i, so its correlation is at most the residue correlation, and usually smaller.

1.4 Results

Every variant is scored on 5000 held-out test sequences, for every k, with two criteria that need no theory:

- **calibration gap** $\mathbb{E}\sum_c Q(c \mid x_U)^2 - \mathbb{E}\,Q(\text{true } c \mid x_U)$. It is 0 for the true posterior (proof in §2.1), positive if Q is over-confident, negative if under-confident;
- **log-loss** $\mathbb{E}[-\log Q(\text{true } c \mid x_U)]$: lower is better.

level	weighting	log-loss	mean |gap|	k_S
family	none (independent sites)	**0.071**	**0.001**	2.1
	geometric mean of $(1-\rho)$	0.073	0.004	3.0
	effective number, ρ^2	0.076	0.007	2.3
	best uniform weight ($w = 0.95$)	**0.071**	**0.001**	2.2
subfamily	none	0.360	0.019	4.9
	geometric mean of $(1-\rho)$	**0.340**	**0.012**	6.5
	effective number, ρ^2	0.367	0.041	6.9
	best uniform weight ($w = 0.70$)	0.343	0.013	6.9
clade	none	0.809	0.065	7.4
	geometric mean of $(1-\rho)$	0.733	0.043	9.7
	effective number, ρ^2	**0.708**	0.044	13.3
	best uniform weight ($w = 0.55$)	0.716	**0.032**	13.4

The product and the effective number with ρ itself fail at every level. For clades, their log-loss is 2.6 and 1.0 and their commitment never reaches $1/2$. With ρ^2 the product also fails (clade log-loss 1.28).

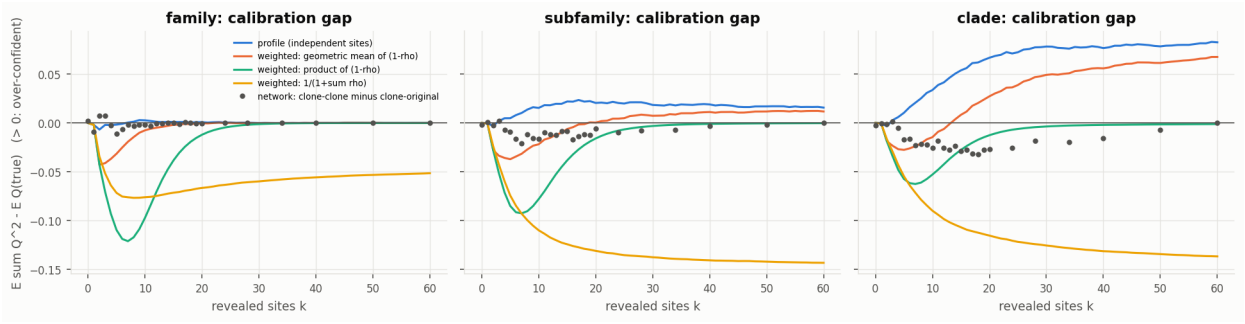


Figure 1: Calibration gap against the number of revealed sites (positive: over-confident). Blue: independent-site profile. Red: your geometric-mean weighting. Green and yellow: product and effective number with ρ. Dots: the network (§2).

What we learn.

1. **Within a class, almost every pair of sites is correlated**: 1730–1770 of the 1770 pairs pass the 5σ threshold, with mean $\rho \approx 0.23$–0.28. The couplings live on only 179 contacts, but they propagate: i–j and j–k in contact make i and k correlated. At the family and subfamily

levels the class is also a mixture of sub-classes, which correlates everything. So every site has almost every other revealed site as a partner.

2. **Your weighting therefore acts as a nearly uniform discount**: the average weight is 0.71 (family), 0.75 (subfamily) and 0.77 (clade), almost independent of k. That is too strong when few sites are revealed, since one partner already costs 25%, and too weak when many are, since the discount does not grow. Fig. 1 (red) shows exactly this: under-confident for $k \lesssim 10$, over-confident beyond.
3. **The formulas whose discount grows with the number of partners over-correct**, because $\sum_j \rho_{ij}$ over $\sim k$ partners of $\rho \approx 0.25$ quickly exceeds 1. The residue correlation over-states the evidence correlation (the caveat of §1.3). Using ρ^2 helps the effective number at the clade level but hurts it at the subfamily level.
4. **No single-site weighting is calibrated everywhere.** The best constant weight leaves a clade gap of 0.03. The rest of the error is not double counting: the profile also ignores the *differences in couplings* between clades, which carry class information that no re-weighting of site terms can recover.
5. **Overall, your geometric mean is the best simple variant.** It is best at the subfamily level and much better than no weighting at the clade level (gap −34%, log-loss −9%). It is slightly worse only at the family level, where the profile was already calibrated.

2 What "under-confident" means

2.1 Definitions and the test

Fix a partial sequence x_U. Two distributions over the class y of its completion matter:

- $P(y \mid x_U)$, the **truth**: the class distribution of true sequences that agree with x_U;
- $Q(y \mid x_U)$, the **network's**: the class distribution of the network's completions of x_U.

We measure two agreements, starting from masked *true* sequences x_0:

$$\text{clone–clone:} \quad \Pr[\text{two clones share the class}] = \mathbb{E}_{x_U} \sum_y Q(y \mid x_U)^2,$$

$$\text{clone–original:} \quad \Pr[\text{a clone shares the class of } x_0] = \mathbb{E}_{x_U} \sum_y Q(y \mid x_U)\, P(y \mid x_U).$$

The second line holds because, given x_U, the original x_0 is itself a draw from the truth, independent of the clones. If $Q = P$ the two lines are equal. Their difference is

$$\text{clone–original} - \text{clone–clone} = \mathbb{E} \sum_y Q(y)\big[P(y) - Q(y)\big].$$

2.2 A one-parameter model: shrinkage towards uniform

Suppose the network's distribution is the truth pulled towards the uniform distribution $u(y) = 1/K$:

$$Q = \lambda P + (1 - \lambda)u.$$

Then $P - Q = (1 - \lambda)(P - u)$. Using $\sum_y u(y)(P(y) - u(y)) = 0$:

$$\sum_y Q(P - Q) = (1 - \lambda)\Big[\lambda \sum_y P(P - u) + (1 - \lambda) \sum_y u(P - u)\Big] = \lambda(1 - \lambda)\Big(\sum_y P^2 - \tfrac{1}{K}\Big).$$

Since $\sum_y P^2 \geq 1/K$, this is **positive for** $0 < \lambda < 1$. That is **under-confidence**: Q is flatter than P, and a clone agrees with the original more often than with another clone. It is **negative for** $\lambda > 1$: over-confidence, Q sharper than P.

Example. $P = (0.8,\ 0.2)$ and $\lambda = 0.5$ give $Q = (0.65,\ 0.35)$. Clone–clone is $0.65^2 + 0.35^2 = 0.545$ and clone–original is $0.8 \times 0.65 + 0.2 \times 0.35 = 0.590$. The difference 0.045 equals $0.5 \times 0.5 \times (0.68 - 0.5)$.

Estimating λ and the true commitment. The model also gives $\sum Q^2 - \frac{1}{K} = \lambda^2\big(\sum P^2 - \frac{1}{K}\big)$ and $\sum PQ - \frac{1}{K} = \lambda\big(\sum P^2 - \frac{1}{K}\big)$. Dividing and squaring:

$$\lambda = \frac{\mathrm{cc} - 1/K}{\mathrm{co} - 1/K}, \qquad \sum_y P^2 = \frac{1}{K} + \frac{(\mathrm{co} - 1/K)^2}{\mathrm{cc} - 1/K},$$

where cc and co are the measured clone–clone and clone–original agreements. This gives an estimate of the *true* commitment curve from the network's clones, corrected for its under-confidence. (It treats λ as the same for all x_U, so it is an approximation.)

2.3 Measured (exact soft labels, §3)

λ at $k =$	2	5	10	20
family	1.02	0.98	1.00	1.00
subfamily	0.99	0.98	0.97	0.99
clade	0.99	0.94	0.95	0.96

clade, $k =$	5	10	20
clone–clone	0.330	0.501	0.695
clone–original	0.347	0.527	0.722

What it means. From a partial sequence, the network's completions land in the *sibling* clade (or subfamily) a little more often than true sequences with the same revealed residues do. It has not extracted all of the clade-specific evidence the revealed residues contain. At the clade level it keeps about 95% of the true confidence. The effect is the same as in Part III (within-family correlations about 10% too weak, per-clade r_C 0.764 against 0.828): the fine, clade-specific signal ($\sigma_h = 0.3$, $\sigma_J = 0.15$ in the tree) is learned only partly.

Two further pieces of evidence.

- **The network's own samples are more ambiguous.** The exact posterior of a complete sequence gives $\mathbb{E}\sum_c P(c \mid x)^2 = 0.954$ for true sequences but only 0.868 for network samples (clade level; 0.999 vs 0.982 for subfamilies). Network samples sit more often *between* two sibling clades.
- **The network is self-consistent.** Starting from masked *network samples*, clone–clone and clone–original agree within 0.01 at every level and k. This is expected if the network's conditionals come from a single joint distribution, which a masked-diffusion network does not guarantee. So the gap measured on true sequences is a gap between the network and the truth, not internal inconsistency.

3 Removing the classifier error

3.1 Why the classifier biases the clade level most

Until now every clone was labelled by the clade with the highest pseudo-likelihood. On labelled test sequences this is right for 100% of families, 99.4% of subfamilies but only 91.5% of clades. Near the boundary between two sibling clades, a completion that differs in a few residues can flip the label. That adds noise to clone agreement, lowers the curve, and delays the 1/2-crossing. Sibling clades differ least, so the effect is largest there.

The right label is the **latent** class, the model the sequence was drawn from. For a complete sequence x it is known only in probability:

$$P(c \mid x) = \frac{e^{-E_c(x)}/Z_c}{\sum_{c'} e^{-E_{c'}(x)}/Z_{c'}} \qquad \text{(equal priors)}.$$

This needs the partition functions Z_c, which the pseudo-likelihood avoided.

3.2 Partition functions by thermodynamic integration

Introduce a coupling strength $\lambda \in [0,1]$ in each clade model:

$$E_\lambda(x) = -\sum_i h_i(x^i) - \lambda \sum_{i<j} J_{ij}(x^i, x^j).$$

(This λ is unrelated to the shrinkage factor of §2.)

1. At $\lambda = 0$ the sites are independent, so $\log Z_0 = \sum_i \log \sum_a e^{h_i(a)}$ exactly.
2. Differentiating $\log Z_\lambda = \log \sum_x e^{-E_\lambda(x)}$:

$$\frac{d \log Z_\lambda}{d\lambda} = \frac{\sum_x \left(\sum_{i<j} J_{ij}\right) e^{-E_\lambda(x)}}{Z_\lambda} = \Big\langle \sum_{i<j} J_{ij}(x^i, x^j) \Big\rangle_\lambda,$$

   an average that Gibbs sampling at strength λ estimates.
3. Integrating: $\log Z_1 = \log Z_0 + \int_0^1 \big\langle \sum_{i<j} J_{ij} \big\rangle_\lambda \, d\lambda$, with Simpson's rule on 21 values of λ and 1000 GPU chains warm-started from one λ to the next.
4. The integral is computed going up and going down in λ. Half their difference estimates the error: at most 0.095 nats over the 18 clades. A 5× longer run on clade 0 gives $\log Z = 340.07$ against 340.10.

`test_18` checks this against exact enumeration on small models (error < 0.02).

Validation on the labelled test set. If the Z_c are right, the posterior is calibrated: $\mathbb{E}\sum_c P^2 = \mathbb{E}P(\text{true } c)$, because the true labels are draws from it.

	family	subfamily	clade
accuracy, exact posterior (Bayes-optimal)	1.000	1.000	**0.969**
accuracy, pseudo-likelihood classifier	1.000	0.994	0.915
$\mathbb{E}\sum_c P(c \mid x)^2$	1.000	0.999	0.954
$\mathbb{E}\,P(\text{true } c \mid x)$	1.000	0.999	0.954

The last two rows agree to 3 decimals, so the Z_c are right. Even the best possible classifier is wrong on 3.1% of clades: sibling clades genuinely overlap, and a complete sequence leaves clade commitment 0.954, not 1.

3.3 Soft-label cloning

Label each clone a by its whole posterior vector $R_a(y) = P(y \mid \text{clone } a)$ instead of a hard label, and measure, for C clones of the same x_U,

$$\widehat{\phi} = \frac{1}{C(C-1)} \sum_{a \neq b} \sum_y R_a(y) R_b(y).$$

Why this is exact. With the exact denoiser, a clone is a draw from $P(x_0 \mid x_U)$, so $\mathbb{E}[R_a(y) \mid x_U] = \sum_{x_0} P(x_0 \mid x_U) P(y \mid x_0) = P(y \mid x_U)$, the latent-class posterior. Two clones are independent given x_U, so $\mathbb{E}[\sum_y R_a R_b \mid x_U] = \sum_y P(y \mid x_U)^2$. No classifier is involved.

At $k = L$ the curve ends at the ceiling $\mathbb{E}\sum_y P(y \mid x)^2$ (0.954 for clades), not at 1. The normalised commitment is therefore $\psi = (\phi - \phi_0)/(\phi_{\text{ceiling}} - \phi_0)$, and k_S is where it crosses 1/2.

3.4 Results

	from held-out true x_t		from the network's own paths	
k_S	classifier labels	exact soft labels	classifier labels	exact soft labels
family	1.93	1.93	1.95	1.94
subfamily	5.83	5.36	5.50	4.91
clade	15.7	**10.1**	17.7	**8.7**

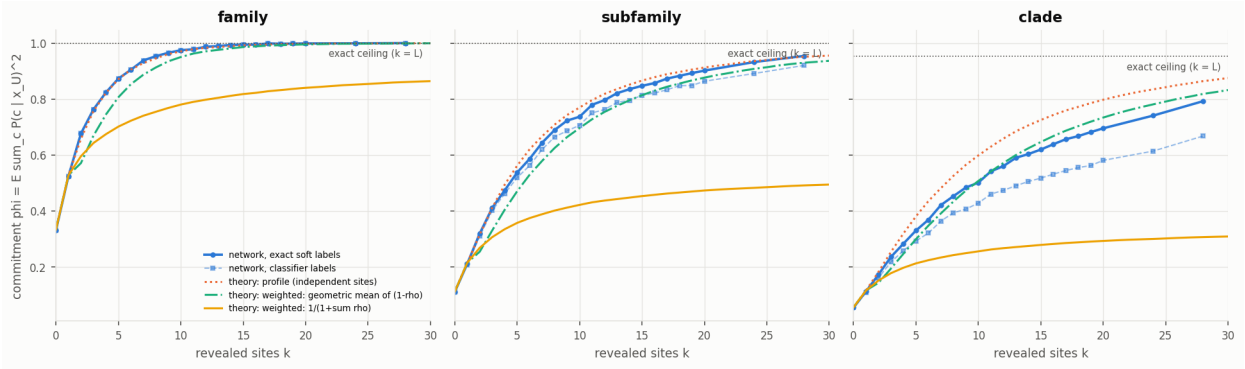


Figure 2: Commitment $\phi = \mathbb{E}\sum_c P(c \mid x_U)^2$ (not normalised). Dark blue: the network with exact soft labels. Light blue: the same clones with classifier labels. Dotted, dash-dotted, solid: profile theory without weighting, with your geometric-mean weighting, and with the effective number. Grey: the exact ceiling at $k = L$.

Classifier noise was responsible for most of the apparent clade delay. With soft labels, the network's clade curve lies between the unweighted profile theory (over-confident) and your weighted one, and it follows the weighted theory closely up to $k \approx 12$ (Fig. 2).

4 Updated speciation points

k_S (revealed sites)	family	subfamily	clade
profile theory, independent sites (over-confident)	2.1	4.9	7.4
profile theory, your geometric-mean weighting	3.0	6.5	9.7
profile theory, best uniform weight (calibrated on average)	2.2	6.9	13.4
network, classifier labels (previous result)	1.9	5.8	15.7
network, exact soft labels	1.9	5.4	10.1
network, soft labels, corrected for under-confidence (§2.2)	**2.0**	**5.0**	**8.2**

How to read the range:

- A theory that is calibrated but uses *less* information than the truth commits *more slowly*. If $Q = \mathbb{E}[P \mid \text{less information}]$, Jensen's inequality gives $\mathbb{E}\sum Q^2 \leq \mathbb{E}\sum P^2$. The calibrated profile (13.4 for clades) ignores the coupling differences between clades, so it is an *upper* bound on the true k_S.
- The network with soft labels is slightly under-confident, so its 10.1 is also an upper bound. Correcting for the shrinkage gives 8.2.
- The independent-site profile is over-confident, so its 7.4 is too early.

Our best estimates are therefore family $\approx$ 2, subfamily $\approx$ 5 and clade $\approx$ 8 revealed sites, i.e. $t_S \approx 0.97,\ 0.92,\ 0.86$. The cascade stands. With these values, the collapse of the empirical denoiser ($k_C = 6.2$, unchanged) still falls between subfamily and clade speciation.